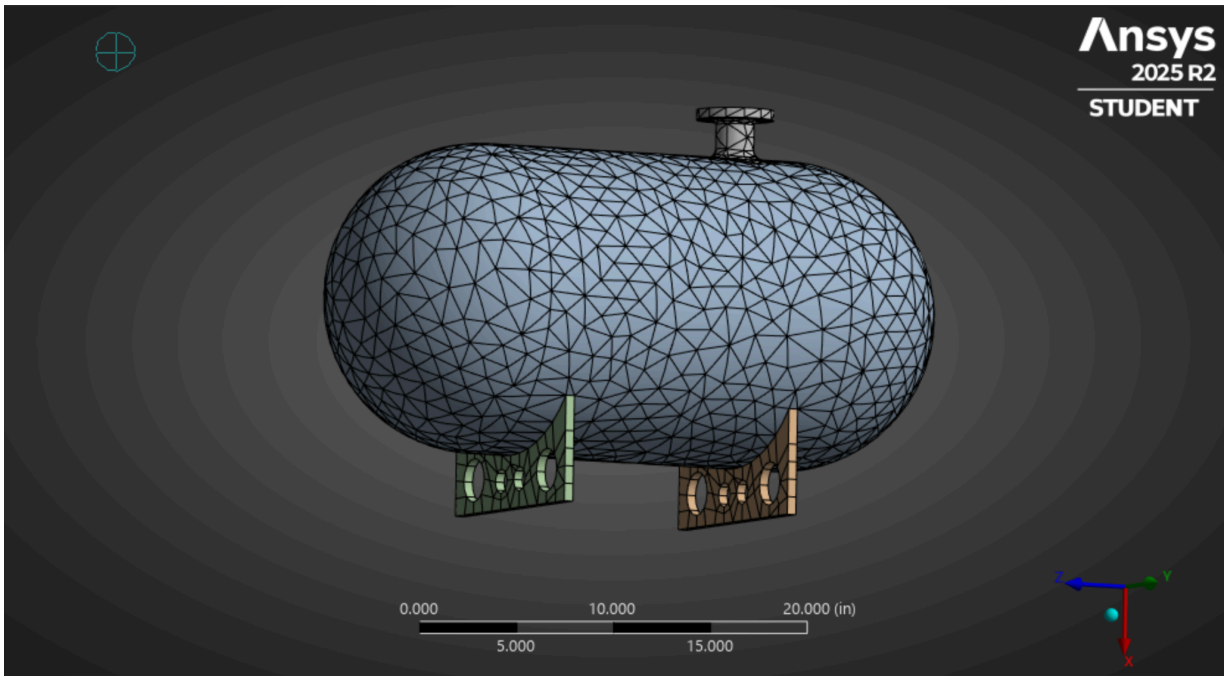
Matthew Heering

Pressure Vessel

3/6/2026

Mesh step



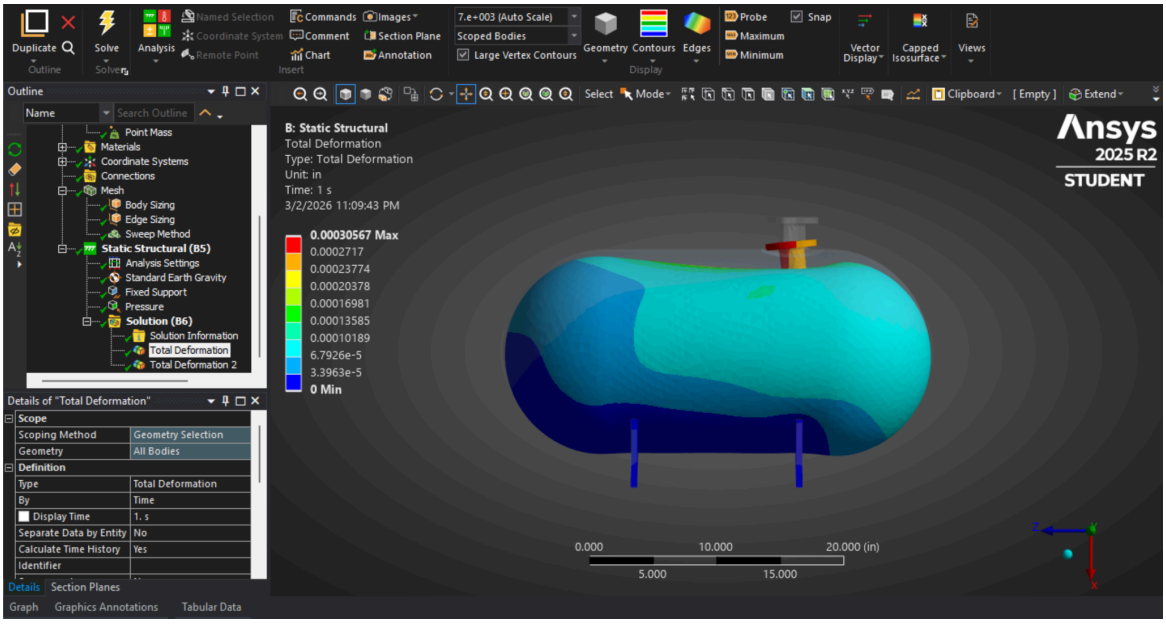
Statistics	
Nodes	13959
Elements	6666

Total number of DOF: 3*13959=41877

The DOF's that can be determined from the essential boundary conditions are the ones associated with the nodes along the bottom surface of the supports. Each node is constrained in all three directions. So for each node on the bottom of the supports, the three degrees of freedom associated with that node are known from the essential boundary conditions.

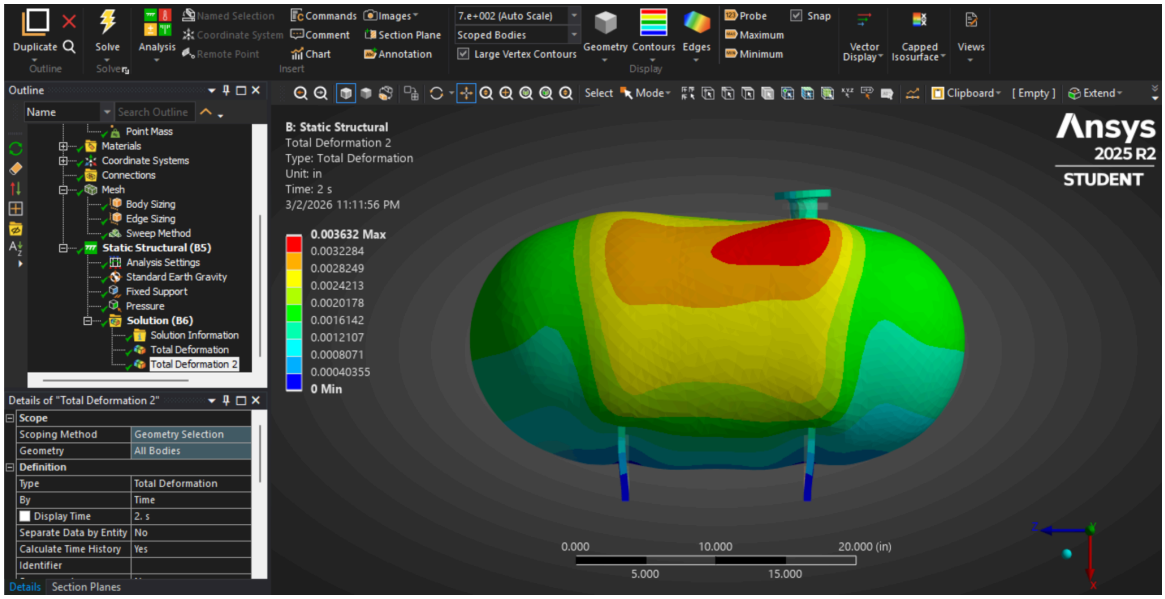
It is modeled as a point mass acting on the top face of the top flange. The magnitude of the traction on the flange if the mass of the valve system is mv is $t = - m_v g / A_f$. This mass is specified in Ansys as a 75lbm point mass on a specified surface.

Displacement magnitude (i.e., “Total Deformation”) when only gravity is applied.



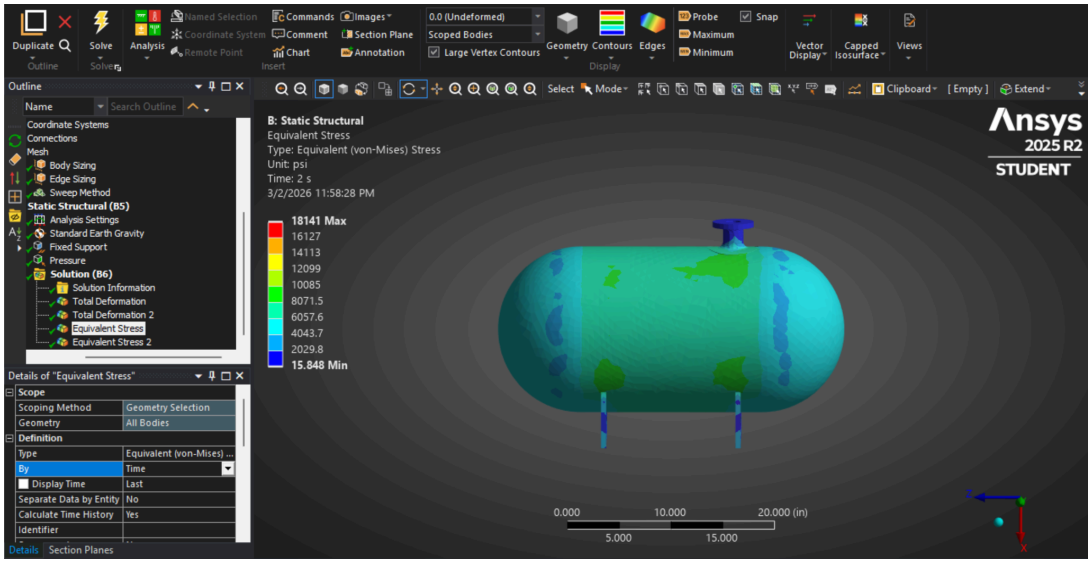
The displacement is plausible since the only non essential applied boundary condition is point mass on the top flange, due to this, the vessel makes sense to collapse inwards in this way from the weight of the valve system.

Displacement magnitude (i.e., “Total Deformation”) when both gravity and pressure are applied.

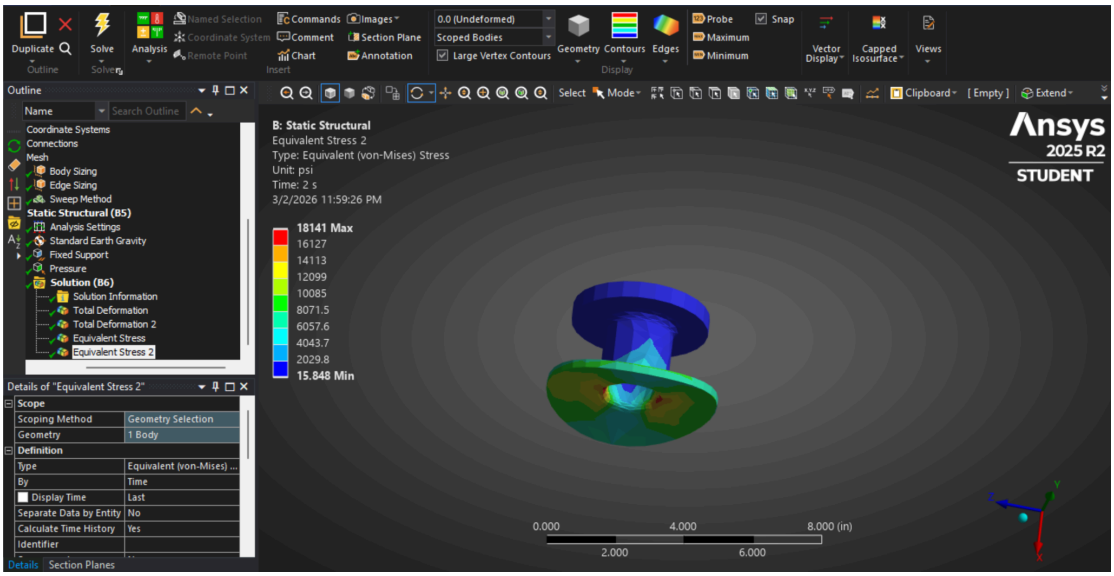


This displacement does seem plausible since the pressure is now applied, and much larger than the force of gravity, so the vessel balloons, and we see minimal collapsing from the weight of the valve.

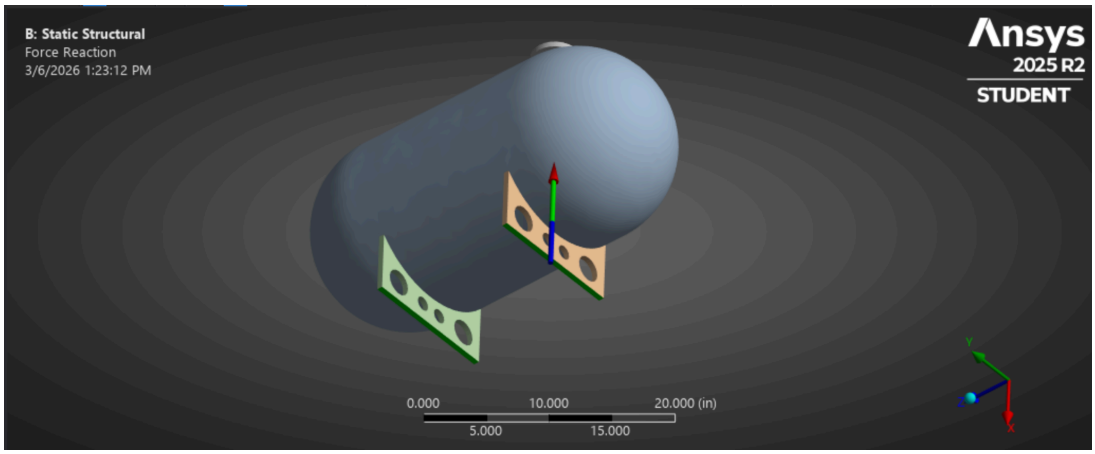
Von Mises stress contours plotted over the undeformed pressure vessel when both gravity and pressure are applied.



Von Mises stress contours plotted over only the nozzle region when both gravity and pressure are applied.

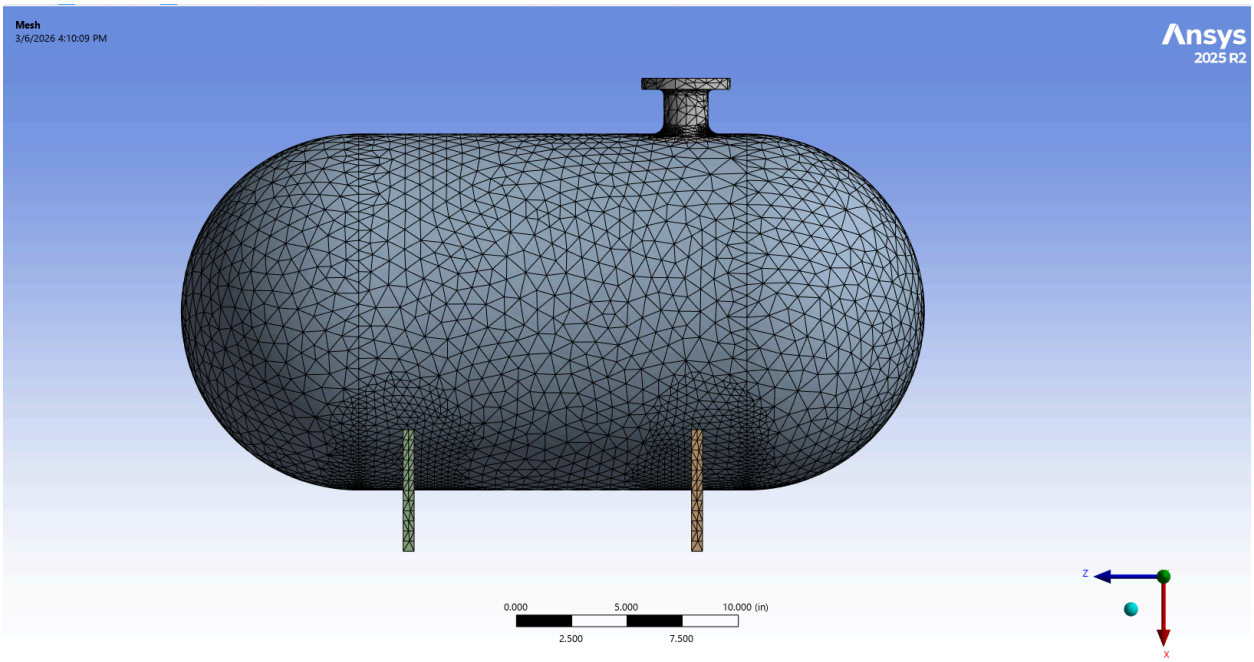


Force Reaction at the bottom of the vessel (where the essential boundary conditions are applied) when only gravity is applied.

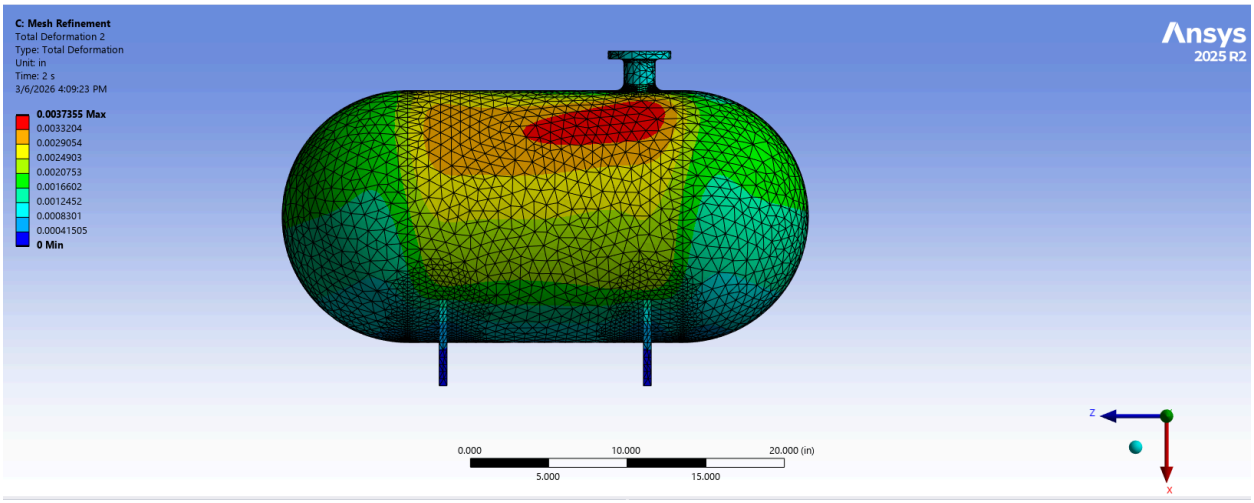


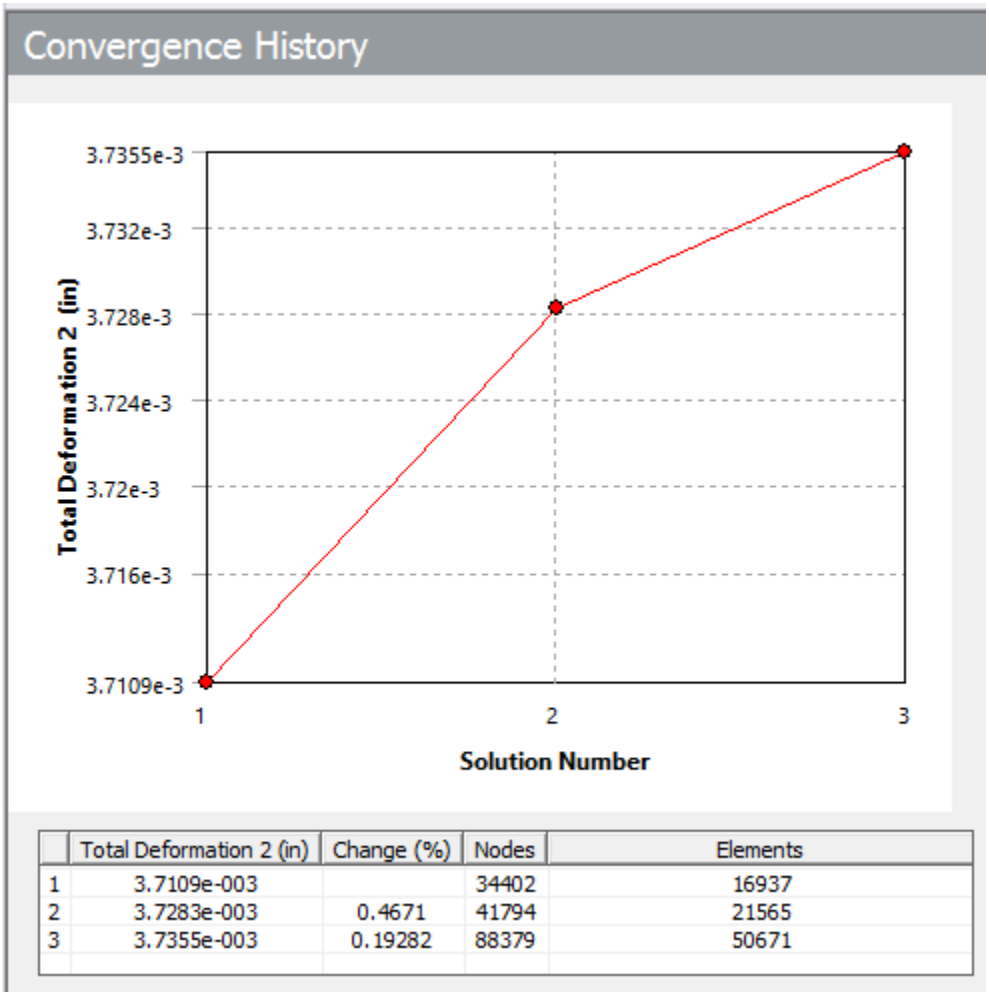
The reaction balances the applied load since it is nearly 715 lbf in +x direction, which we calculated to be the necessary reaction to keep the system in equilibrium. The reaction is calculated by plugging the known displacements from the nodes along the essential boundary condition into the equilibrium equation and solving for the required force to enforce the displacement constraints.

Refined mesh.



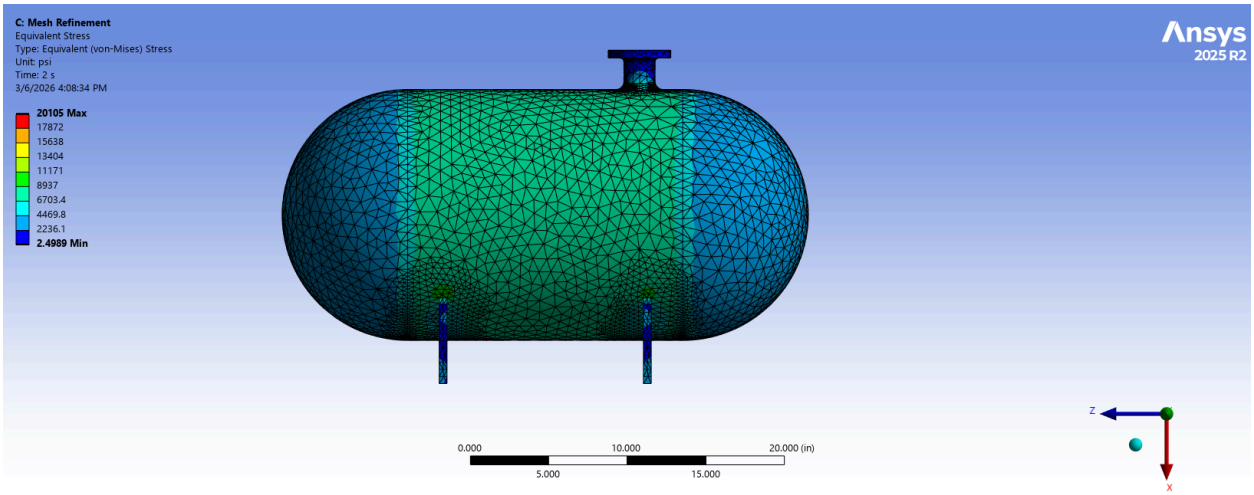
Snapshot of the displacement magnitude contours along with the corresponding convergence graph.

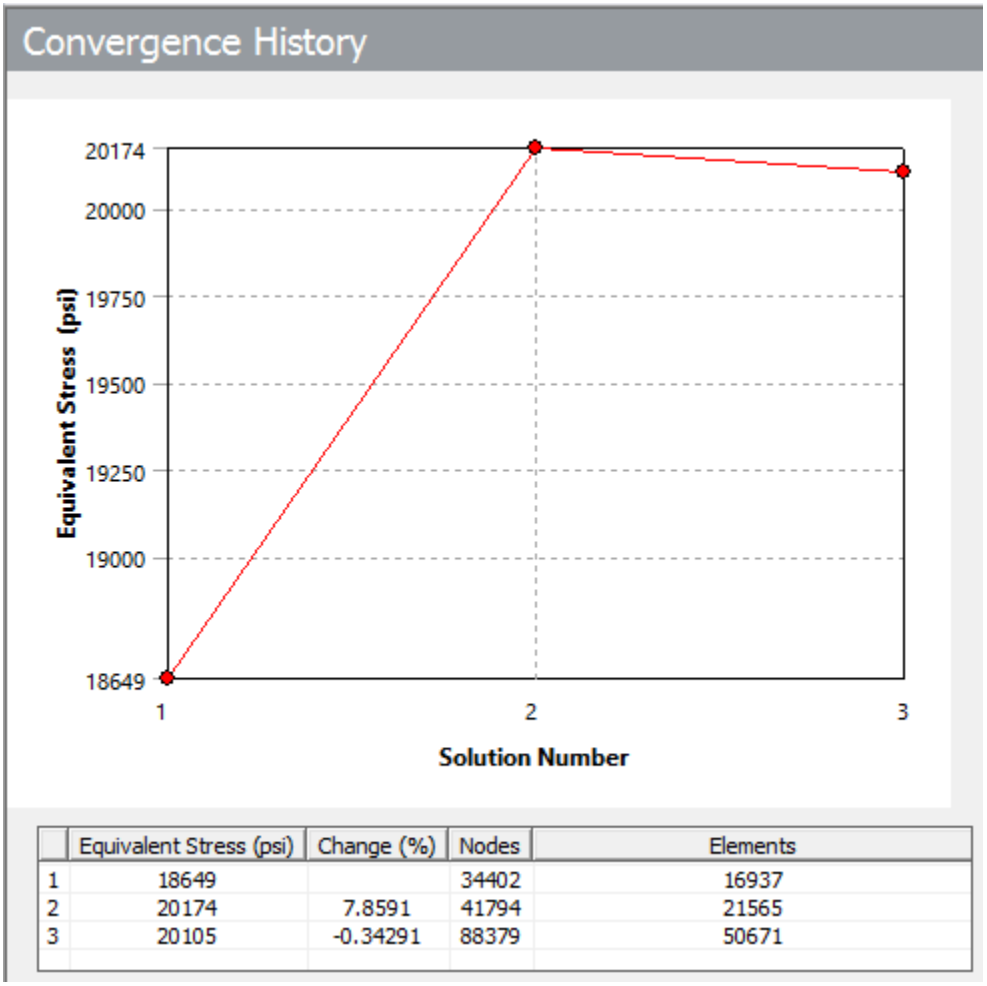




	Total Deformation 2 (in)	Change (%)	Nodes	Elements
1	3.7109e-003		34402	16937
2	3.7283e-003	0.4671	41794	21565
3	3.7355e-003	0.19282	88379	50671

Snapshot of the von Mises stress contours along with the corresponding convergence graph.





	Equivalent Stress (psi)	Change (%)	Nodes	Elements
1	18649		34402	16937
2	20174	7.8591	41794	21565
3	20105	-0.34291	88379	50671

Von Mises stress is much more sensitive to mesh as we can see from the convergence history graphs. There is a much greater difference in the magnitude of the percent change for equivalent stress than for total deformation 2.